

SMART INDIA HACKATHON 2026 — OFFICIAL PROBLEM STATEMENT

Problem Statement ID:	SIH261211	Organization:	Oil India Limited (OIL)
Track:	Smart Automation	Category:	Software / AI
Project Title:	eRTMAC-NWIS (Nearby Wells Intelligence System)		Drilling Engineering

PROBLEM STATEMENT DESCRIPTION:

Oil India Limited (OIL) routinely drills deep exploratory and development wells in the Upper Assam Shelf Basin. Drilling engineers encounter high-risk subsurface hazards including Stuck Pipe in Barail shales, Gas Kicks in Kopili overpressures, and Mud Loss in Sylhet karst limestones, causing multi-crore Non-Productive Time (NPT) losses (■10L to ■50L/day rig costs). While historical offset wells encountered identical hazards and documented solutions in scanned Well Completion Reports (WCRs) and Daily Drilling Reports (DDRs), this critical knowledge remains trapped in legacy PDF archives. The eRTMAC-NWIS platform solves this by providing a 4-Dashboard AI Decision Support System integrating GIS Offset Selection, Pre-Drill Risk Heatmaps, Live Telemetry Anomaly Detection, and RAG PDF Document Knowledge Assistant.